Emergent time scales of epistasis in protein evolution
Supplementary information

1. EVOLUTIONARY ALGORITHM

Sampling from high-dimensional probability distributions usually requires advanced algorithms, such as Markov Chain Monte Carlo (MCMC) sampling. For spin models defined over categorical variables, like Potts models, many sampling algorithms are available. For example, Metropolis-Hastings and Gibbs Monte Carlo are both utilized as a sampling technique in the implementation of Boltzmann learning to infer protein sequence landscapes [1]. In this context, MCMC sampling generates equilibrium sequences via long simulations, ensuring the proper exploration of sequence space. However, MCMC sampling can also be used to model the dynamics of biological systems by properly interpreting the Markov chain configurations [2]. In particular, it was shown that Markov chain sampling can be used to model short-term protein evolution in neutral genetic drift experiments: short simulations performed with Gibbs sampling, with the addition of the constraints of the genetic code, accurately reproduced many statistical features of the experimental data [3]. In the next sections, we discuss how to extend the approach of Refs. [2, 3] to sample long chains that reach equilibrium, thereby modeling protein evolution over billions of years, while also accurately incorporating the amino acid accessibility dictated by the genetic code.

A. Model definition over nucleotide space

As a generative model of amino acid sequences, we consider a Potts model

$$P(\mathbf{a}) = \frac{e^{-\beta H(\mathbf{a})}}{Z} \tag{S1}$$

whose parameters of the Hamiltonian H have been inferred from the natural sequences of a protein family and $\beta = 1/T$ is the inverse temperature. Using this model, we aim to define an evolutionary dynamics taking into account that evolution happens in nucleotide space. This necessitates specifying the probability distribution of $P(\mathbf{a})$ over nucleotide sequences instead of amino acid sequences. A straightforward procedure to accomplish this consists of uniformly distributing the probability associated with an amino acid sequence across all its synonymous nucleotide sequences. To formalize this concept, we define an amino acid sequence of length L as $\mathbf{a} = (a_1, a_2, ..., a_L)$ and a corresponding nucleotide sequence as $\mathbf{n} = (n_{1_1}, n_{1_2}, n_{1_3}, n_{2_1}, ..., n_{L_3})$. The function $A(\mathbf{n})$ applies the genetic code to transform the nucleotide sequence $\mathbf{n}$ into its respective amino acid sequence $\mathbf{a}$. For each amino acid sequence $\mathbf{a}$, we also define the set

$$\boldsymbol{\Gamma}(\mathbf{a}) = \{\mathbf{n} : A(\mathbf{n}) = \mathbf{a}\}, \tag{S2}$$

and the function $N(\mathbf{a}) = |\boldsymbol{\Gamma}(\mathbf{a})|$, i.e. the cardinality of $\boldsymbol{\Gamma}(\mathbf{a})$, which counts the number of synonymous nucleotide sequences coding for a given amino acid sequence $\mathbf{a}$. Because each amino acid is coded independently, $N(\mathbf{a})$ factorizes over the amino acids and can be rewritten as:

$$N(\mathbf{a}) = \prod_{i=1}^{L} N(a_i) \tag{S3}$$

where $N(a)$ is the number of synonymous codons coding for amino acid a. We now have all the elements to define the probability distribution of a new model $\mathcal{P}(\mathbf{n})$ over nucleotide space:

$$\mathcal{P}(\mathbf{n}) = \frac{P(A(\mathbf{n}))}{N(A(\mathbf{n}))} = \frac{1}{N(A(\mathbf{n}))} \frac{e^{-\beta H(A(\mathbf{n}))}}{Z} = \frac{e^{-\beta \mathcal{H}(\mathbf{n})}}{Z} \tag{S4}$$

such that the new Hamiltonian is defined as:

$$\begin{aligned} \mathcal{H}(\mathbf{n}) &= H(A(\mathbf{n})) + T \log\left[N(A(\mathbf{n}))\right] \\ &= H(\mathbf{a}) + T \sum_{i=1}^{L} \log\left[N(a_i)\right]. \end{aligned} \tag{S5}$$

The Hamiltonian in Eq. (S5) incorporates a novel term compared to the standard Potts Hamiltonian defined over amino acid space. Namely,

$$T \sum_{i=1}^{L} \log\left[N(a_i)\right] \tag{S6}$$

which accounts for the entropic contribution arising from the degeneracy of synonymous DNA sequences.

The term in Eq. (S6) introduces a correction that disfavors nucleotide sequences with high genetic redundancy by assigning them higher energy, i.e. lower probability, compared to sequences with low degeneracy which are assigned lower energy, i.e. higher probability. Consequently, the original Potts probability of amino acid sequence **a** defined in Eq. (S1) is correctly recovered after summing over all synonymous nucleotide sequences:

$$P(\mathbf{a}) = \sum_{\mathbf{n}\in\boldsymbol{\Gamma}(\mathbf{a})} \frac{e^{-\beta H(A(\mathbf{n}))}}{Z}. \tag{S7}$$

An alternative approach to define the probability distribution on nucleotide space is to introduce a codon bias, i.e. to assign non-uniform weights to the codons, accounting for the specific codon usages of different species. Let us alternatively denote a nucleotide sequence by a sequence of codons $\mathbf{c} = (c_1, \cdots, c_L)$, which is then reflected at the amino acid level $\mathbf{a}(\mathbf{c})$, using the genetic code. We define also $f(c_i|a_i)$ the frequency with which codon c_i codes for amino acid a_i. From the relation

$$P(\mathbf{c}) = P[\mathbf{a}(\mathbf{c})] \prod_i f(c_i|a_i)\ , \tag{S8}$$

we can introduce a novel nucleotide sequence energy

$$\mathcal{H}(\mathbf{c}) = -T\log P(\mathbf{c}) = E[\mathbf{a}(\mathbf{c})] - T\sum_{i=1}^{L} \log f(c_i|a_i)\ , \tag{S9}$$

If $f(c_i|a_i) = 1/N(a_i)$ we fall back in the previous description without codon bias, but one can consider different choices, e.g. by estimating $f(c_i|a_i)$ from the codon bias of the species in which a given experiment is performed. However, for long term evolution, it does not make sense to consider the species-dependent codon bias, as natural evolution takes place across different species. In Fig. S9 we show the codon usage measured from in silico sequences generated via simulations with or without codon bias. In our case, introducing a codon bias for the experiments at short time scales is not relevant for modeling purposes, because the mutational process was performed in vitro, hence we assume no codon bias to carry over our analysis.

B. Description of the algorithm

Detailed balance (DB) is a mathematical condition ensuring that the configurations generated by a Markov process converge to a stationary distribution. In our case, DB ensures that the Markov chains converge to the statistics of natural amino acid sequences that we used to infer the Potts model. Unfortunately, the sampling dynamics on nucleotides introduced in Ref. [3] does not respect the DB condition. Adapting the sampling algorithm on nucleotides to satisfy DB is the goal of this section. DB is a statement about the transition probabilities π between configurations of the Markov Chain. In the case of nucleotide sequences, for every pair of sequences $\mathbf{n}, \mathbf{n}'$, the following expression must hold:

$$\mathcal{P}(\mathbf{n})\pi(\mathbf{n} \to \mathbf{n}') = \mathcal{P}(\mathbf{n}')\pi(\mathbf{n}' \to \mathbf{n})\ . \tag{S10}$$

The condition requires that, at equilibrium, the probability flux $\mathbf{n} \to \mathbf{n}'$ equals the one from $\mathbf{n}' \to \mathbf{n}$, i.e. the probabilities of $\mathbf{n}$ and $\mathbf{n}'$ do not change. A vast class of Markov Chain Monte Carlo algorithms depends on finding a suitable transition matrix that satisfies this equation. In addition to that, we want π to model a plausible evolutionary dynamics on nucleotides. To this aim, we need to account for at least three processes:

- single-nucleotide mutations: a nucleotide is replaced by another one
- amino acid deletion: a codon is lost from the sequence, removing 3 nucleotides

- amino acid insertion: a codon is inserted in the sequence, adding 3 nucleotides

In terms of indels, we model only triplets on nucleotides because inserting or deleting single (or pairs of) bases results in a frameshift, misaligning subsequent codons in the reading frame and leading with very high probability to non-functional sequences. On top of this, in terms of the amino acid sequence, this move is highly non-local, thus hard to model. As a consequence, we assume that this process has zero probability and we exclude it from the dynamics. While this approach represents a simplification of the actual indel statistics, we contend that modeling triplets of indel maintains a biological foundation. Replication slippage, one of the most well-understood mechanisms generating indels in protein evolution, exemplifies this biological basis. During DNA replication, the DNA polymerase sometimes pauses and disconnects from the DNA. This pause can allow the new, growing DNA strand to momentarily detach and then accidentally reconnect to a similar sequence either ahead or behind the original spot. When the DNA polymerase restarts copying, it may either miss or repeat a section of the DNA, creating a deletion or an insertion in the new DNA strand.

Including both indel mutations and single-nucleotide substitutions in a single MCMC framework is not trivial, since they act differently on variables and satisfaction of DB cannot be guaranteed. To overcome this issue, we suggest an approach that combines Gibbs and Metropolis sampling. We devised a mixed sampler operating as follows:

- with probability p, execute a Metropolis move simulating codon indels, i.e. operating on triplets of nucleotides.
- with probability $1 - p$, perform a Gibbs move simulating a single-nucleotide mutation.

Note that this also reflects the different nature of the two processes: our dynamics determines first which process happens, and then its outcome.

B.1. Indels: Metropolis sampling

To model codon insertions and deletions (indels), we introduce a new symbol: the triple gap "---" which we denote with c^0. This codon, equivalent to an amino acid gap, models the deletion of a codon within a nucleotide sequence. We also designate with c^k, $k \in \{1, ..., 64\}$, all other amino-acid-coding codons, including the stop codons.

We can now we decompose the Metropolis transition matrix $\pi(\mathbf{n} \to \mathbf{n}')$ into two components: a proposal term $p(\mathbf{n} \to \mathbf{n}')$ and an acceptance term $\alpha(\mathbf{n} \to \mathbf{n}')$. Since we exclusively focus on single codon substitutions, we only need to consider transitions $\mathbf{n} \to \mathbf{n}'$ such that in codon language $\mathbf{n} = \mathbf{c} = (c_1, c_2, \ldots, c_i, \ldots, c_L)$ and $\mathbf{n}' = \mathbf{c}' = (c_1, c_2, \ldots, c_i', \ldots, c_L)$. As a consequence, we can focus on the single-codon transition matrices $p(c \to c')$ and $\alpha(c \to c')$.

We restrict our proposal to mutations from amino-acid-coding codons towards the gap codon and vice versa, leading to the following proposal matrix $p(c \to c')$ where the missing elements are set to zero:

$$\begin{array}{c|ccccc} & c^0 & c^1 & c^2 & \cdots & c^{64} \\ \hline c^0 & \eta & \beta & \beta & \cdots & \beta \\ c^1 & \beta & \gamma & & 0 & \\ c^2 & \beta & & \gamma & & \\ \vdots & \vdots & 0 & & \ddots & \\ c^{64} & \beta & & & & \gamma \end{array} \tag{S11}$$

This symmetric matrix accommodates insertions and deletions via the parameter β and prohibits amino acid substitutions:

$$p(c^m \to c^n) = 0 \iff m \neq n \quad \text{or} \quad m, n > 0. \tag{S12}$$

The parameters η and γ are used to normalize the proposal probability and correspond to "empty" moves not changing the state of the codon. To speed up the algorithm, we want to maximize β while maintaining the proposal probability normalized. As a result we get $\eta = 0$, $\beta = 1/64$ and $\gamma = 63/64$. This means that when we select an amino-acid-coding codon, in 63 out of 64 attempts we propose the same codon, i.e. nothing changes, and only in one case we do propose the triple gap. If a stop codon is proposed, it never gets accepted.

We are now in a position to describe how the Metropolis sampling algorithm for indels works. For each Monte Carlo step:

1. A sequence position $i \in \{1, \dots, L\}$ is randomly selected, along with its corresponding c_i codon.

2. A new codon c_i' is proposed though the proposal matrix $p(c_i \to c_i')$

3. The acceptance probability of the proposed transition $c_i \to c_i'$ is computed according to the Metropolis prescription:

$$\alpha(c \to c') = min\left(1, \frac{\mathcal{P}(\mathbf{n}')}{\mathcal{P}(\mathbf{n})}\right) = min\left(1, e^{-\beta[\mathcal{H}(\mathbf{n}') - \mathcal{H}(\mathbf{n})]}\right) \tag{S13}$$

4. Codon c_i' is accepted with probability $\alpha(c_i \to c_i')$ or refused with probability $1 - \alpha(c_i \to c_i')$.

Thanks to the fact that the proposal matrix in Eq. (S11) is symmetric and that we accept codon mutations according to Eq. (S13), detailed balance is guaranteed.

We highlight that insertions and deletions are a crucial feature of the algorithm to converge to the correct stationary probability distribution and not just a detail that has been introduced for completeness. Despite having a lower statistical energy, sequences sampled without indels do not correctly cover the main two principal directions of natural data in PCA space and they have a lower diversity/pairwise hamming with respect to natural data (Fig. S10).

B.2. Point mutations: Gibbs sampling

The second type of mutation that we need to consider is single-nucleotide substitutions. We resort to Gibbs Monte Carlo sampling as in Ref. [3], but we generalize that approach by introducing DB for accurate long-term sampling. Gibbs sampling works by iteratively emitting new variables from their conditional distribution, given the current value of the rest of the sequence. According to Eq. (S4), we can express the conditional probability of n_{i_k} (i.e. nucleotide n belonging to the i-th codon, position k) given the rest of the sequence as:

$$\begin{aligned} \mathcal{P}(n_{i_k}|\mathbf{n}_{-i_k}) &= \frac{\mathcal{P}(\mathbf{n})}{\mathcal{P}(\mathbf{n}_{-i_k})} = \frac{\mathcal{P}(\mathbf{n})}{\sum\limits_{n_{i_k} \in \mathcal{N}} \mathcal{P}(\mathbf{n})} \\ &= \frac{e^{-\beta\mathcal{H}(n_{1_1}, n_{1_2}, \dots, n_{i_k}, \dots, n_{L_3})}}{\sum\limits_{n \in \mathcal{N}} e^{-\beta\mathcal{H}(n_{1_1}, n_{1_2}, \dots, n_{i_k} = n, \dots, n_{L_3})}} \end{aligned} \tag{S14}$$

where $\mathcal{N}$ = {A, C, G, T}.

We can now present how a step of Gibbs Monte Carlo sampling works in our algorithm:

1. A nucleotide position i_k, $i \in \{1, \dots, L\}$ and $k \in \{1, 2, 3\}$, is randomly selected, excluding currently gapped positions.

2. In order to explicitly compute the value of Eq. (S14), the set of amino acids $\mathcal{A}_{i_k}$ is generated, which is the translation of the 4 possible codons in position i derived by inserting each of the nucleotides of $\mathcal{N}$ in position i_k.

3. A new amino acid n', and the respective amino acid a', is sampled from the following probability distribution, computed exploiting the form of the Hamiltonian in Eq. (S5):

$$\begin{aligned} \mathcal{P}(n_{i_k} = n'|\mathbf{n}_{-i_k}) &= \frac{e^{-\beta H(a_1, \dots, a_i = a', \dots, a_L) - \log[N(a_1, \dots, a_i = a', \dots, a_L)]}}{\sum\limits_{a \in \mathcal{A}_{i_k}} e^{-\beta H(a_1, \dots, a_i = a, \dots, a_L) - \log[N(a_1, \dots, a_i = a, \dots, a_L)]}} \\ &= \frac{e^{\beta h_i(a') + \beta \sum_j J_{ij}(a', a_j) - \log[N(a')]}}{\sum\limits_{a \in \mathcal{A}_{i_k}} e^{\beta h_i(a) + \beta \sum_j J_{ij}(a, a_j) - \log[N(a)]}} \end{aligned} \tag{S15}$$

4. n' is accepted unless it produces a stop codon. In either case, the procedure is restarted from point 1.

Eq. (S15) is quick to evaluate, thanks to the many simplifications happening. In particular, the fields h and the couplings J between unmutated sites, as well as the degeneracies $N(\mathbf{n})$ of unmutated codons appear in both the numerator and the denominator, so they can be divided out from the expression, leaving only parameters coupled to the mutated site.

C. Comparison with previous algorithms

The evolutionary model that has been thoroughly explained in the previous sections was developed by iteratively taking information from previous models. In fact, a first important contribution highlighted how Gibbs sampling Monte Carlo dynamics on epistatic landscape models [2] reproduces several statistical properties of protein evolution, while being consistent with neutral evolution theory, unifying observations from previously distinct evolutionary sequence models. A second contribution has applied this idea to modeling experimental protein evolution, which can only reach up to the short-term scale of $\simeq 20\%$ of divergence from the wildtype [4, 5]. It was shown that the inclusion of genetic code constraints on mutations leads to higher correlations with experimental data [3]. Despite these advances, the results of Ref. [3] were confined to (i) short timescales and (ii) sequences that do not contain gaps. In particular, because the selection procedure of Ref. [3] did not satisfy detailed balance, the resulting dynamical model was not suited to reproduce the long-term equilibrium properties of protein family alignments. To improve upon previous results [2, 3, 6], our new in silico evolutionary dynamics uses a simple, biologically motivated stochastic process that takes into account mutational constraints due to the genetic code, and the possibility of insertions and deletions, while still satisfying detailed balance.

2. SUPPLEMENTARY FIGURES

A. Additional data on β-lactamases

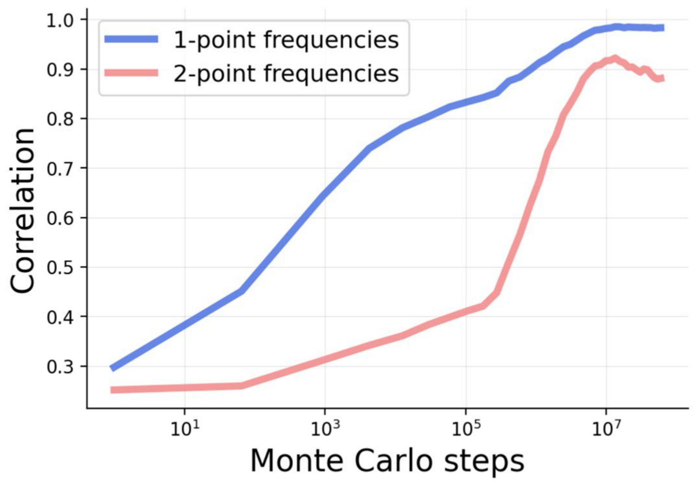

Fig. S1. Statistical properties of 1000 in silico sequences for the β-lactamase family at different times. Pearson correlation of the single-point $f_i(a)$ and connected two-point $c_{ij}(a,b)$ statistics between natural sequences and a set of 1000 in silico sequences sampled over time. The starting MSA is entirely composed of PSE-1 wt sequences. The plot shows how the independent Monte Carlo chains progressively diversify, asymptotically reproducing the statistics of natural sequences.

B. Validation of the model for the DBD family

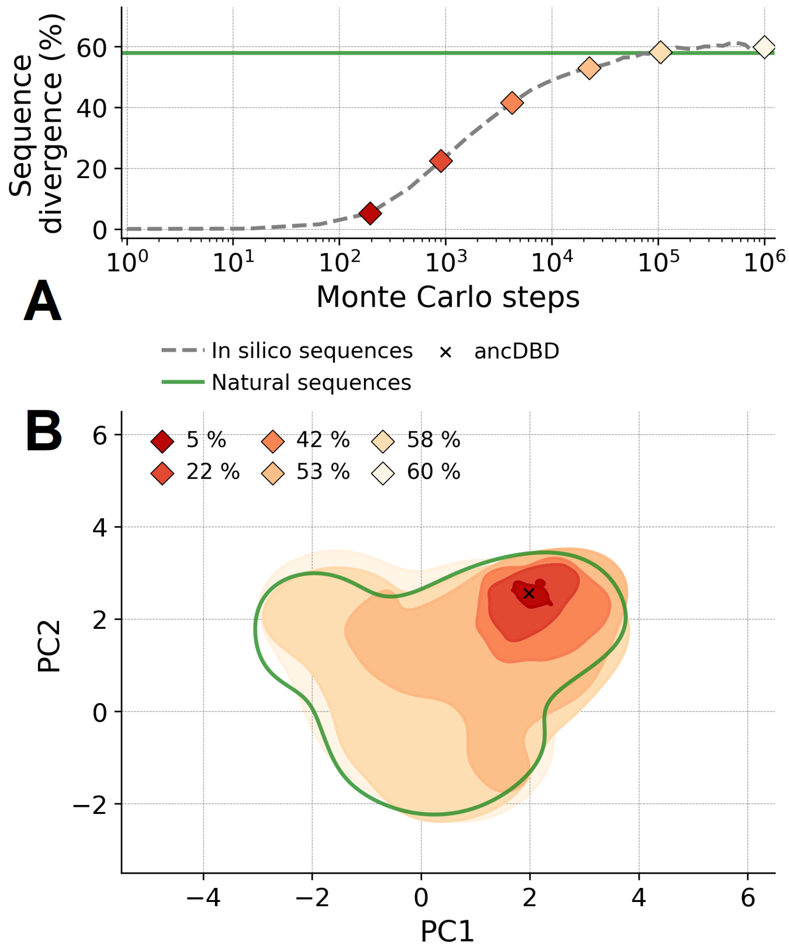

Fig. S2. Illustration and validation of the evolutionary model on the DBD protein family. (A) Average sequence divergence (normalized Hamming distance) from the reconstructed ancestral sequence in which simulation has started over 100 Markov chain realizations of the mutational dynamics at $\beta = 1$. The green line depicts the average sequence divergence of the family of natural proteins used as a training set. (B) Projection over the two principal components (PCA) of the natural sequences (green) and of the sequences sampled after a given number of in silico evolutionary steps (red to light orange, labeled by the average sequence divergence).

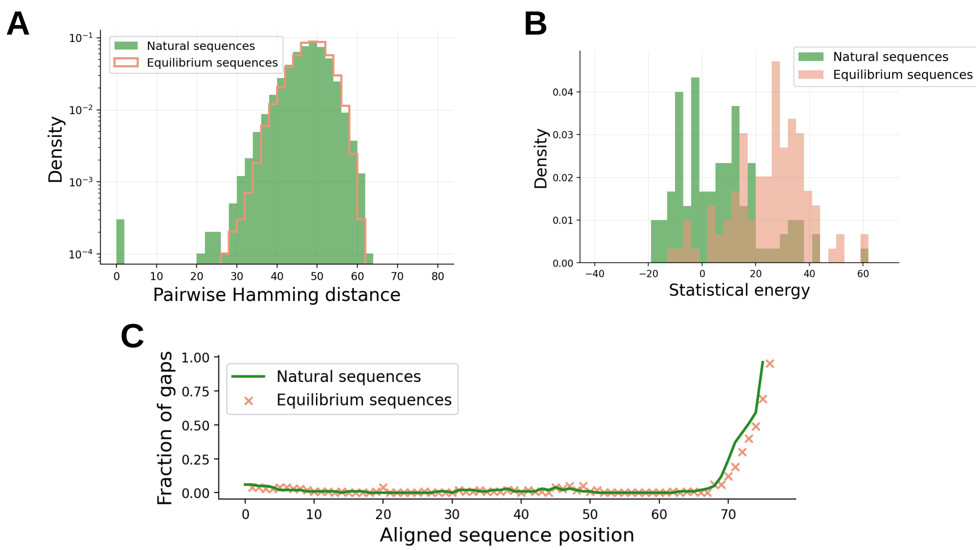

Fig. S3. Statistical properties of 100 equilibrium in silico sequences for the DBD family. (A) Pairwise Hamming distance of natural and equilibrium sequences. (B) Energy of natural and equilibrium sequences. (C) Fraction of gaps per position of natural and equilibrium sequences. The excess of gaps at the extremities of the alignment is due to the alignment procedure.

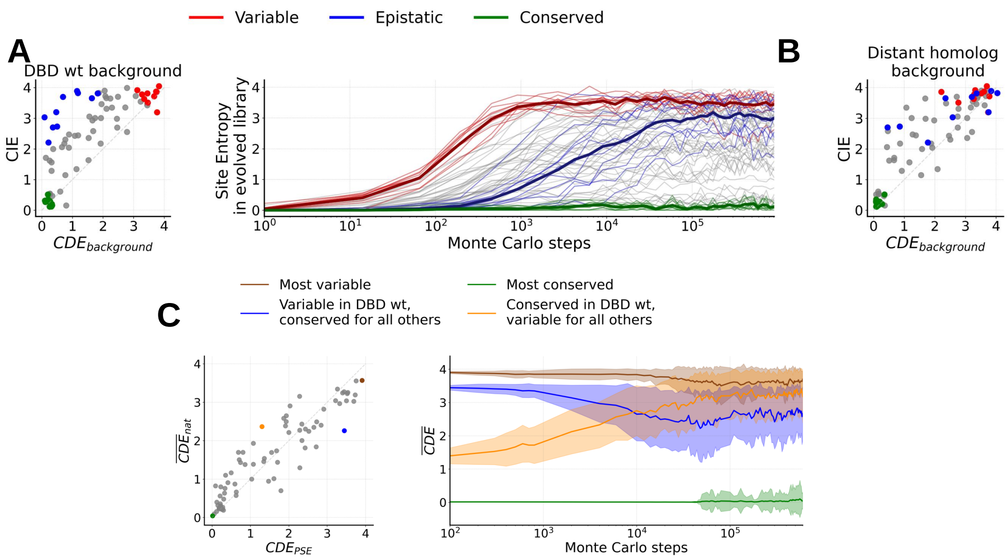

Fig. S4. In silico evolutionary dynamics of variable, conserved, and epistatic sites for the DBD family. (A) Left: classification of residues according to local variability (CDE) computed with the bmDCA model in the ancestral DBD sequence from Ref. [7] (wt) context and global variability (CIE) computed using the amino acid frequencies in natural proteins. We identify three categories of 10 sites each: conserved (green), mutable (red) and epistatic (blue). Right: Site entropy of a library constructed by 100 independent MCMC samplings initialized in the wt sequence, with the three different site categories highlighted; faded lines refer to each site whereas the three thick lines are the mean over the three categories. (B) Same scatter plot as in the left panel of A, but in the context of a distant homolog. The site colors are the same as for the DBD ancestral background, highlighting that site categories are reshuffled in a different context. (C) Left: Illustration of four prototypical sites in the alignment and scatter plot of their CDE computed in the wt context versus the mean CDE on all proteins of the DBD family. Right: CDE of the four sites identified in the left panel, averaged over the context generated by 100 independent MCMC trajectories (thick lines). The faded area indicates the standard deviation.

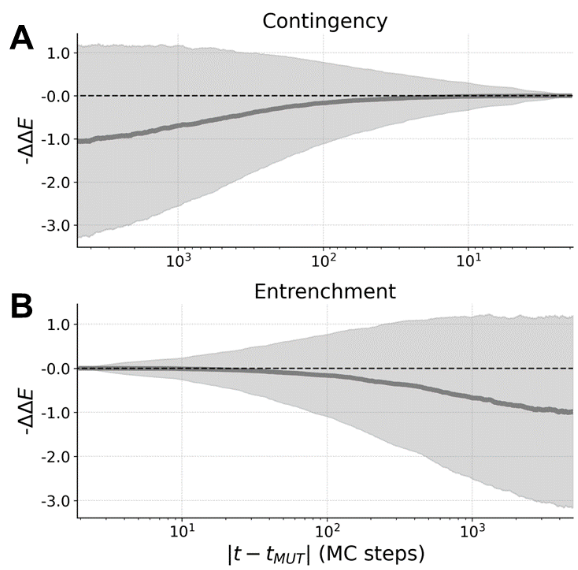

Fig. S5. Contingency and entrenchment from in silico evolution for the DBD family. We select 5000 mutations labeled by *aib* (i.e. site i, aminoacid $a \to b$) along an evolutionary trajectory of the DBD family. (A) Contingency: average value of the fitness gain $-\Delta\Delta E$ between the original mutation *aib* at time t_{MUT}, and the mutational effect of introducing amino acid b in site i over a time window of 5000 MCMC steps before t_{MUT}. The faded area is the standard deviation. (B) Entrenchment: average value of the fitness gain $-\Delta\Delta E$ between the original mutation *aib* at time t_{MUT}, and the mutational effect of reverting to aminoacid a over a time window of 5000 MCMC steps after t_{MUT}. The faded area is the standard deviation.

C. Dynamics of the local entropy in the absence of coevolutionary couplings

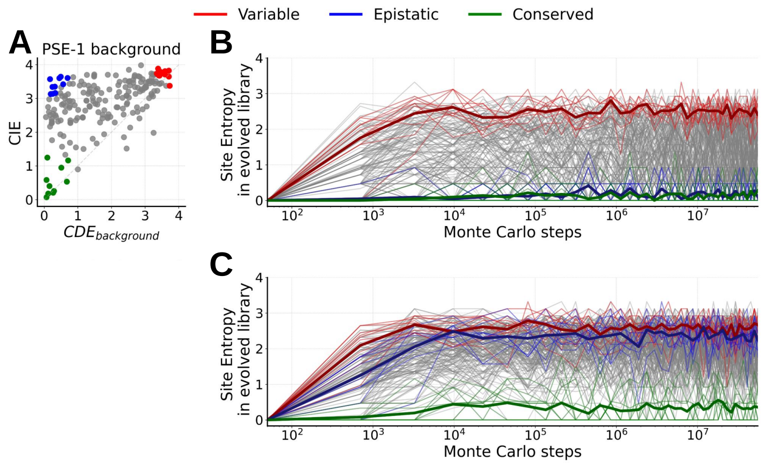

Fig. S6. In silico evolutionary dynamics of variable, conserved, and epistatic sites for the β-lactamase family using profile models. To highlight the importance of the coevolutionary couplings, here we report the same analysis of Fig. 2 of the main text, but for a model without couplings, i.e. a profile model. The labels and color codes are identical to Figs. 2 and S4. Panel B reports the results for a profile model that reproduces the context of PSE-1 as predicted by the bmDCA model, i.e. we set $\pi_i(a_i) = P_{bmDCA}(a_i|\text{PSE-1})$ for each site and the global probability of the profile model is $P(a_1, \cdots, a_N) = \prod_{i=1}^{N} \pi_i(a_i)$. Panel C reports the results for a profile model based on the natural amino acid frequencies, i.e. $\pi_i(a_i) = f_i(a_i)$. As expected, we observe that in the absence of couplings, all site entropies evolve quickly to their equilibrium entropy in the profile model, which is the CDE in the PSE-1 background in panel B, and the CIE in panel C.

D. Classification of sites according to the DMS in the DBD family

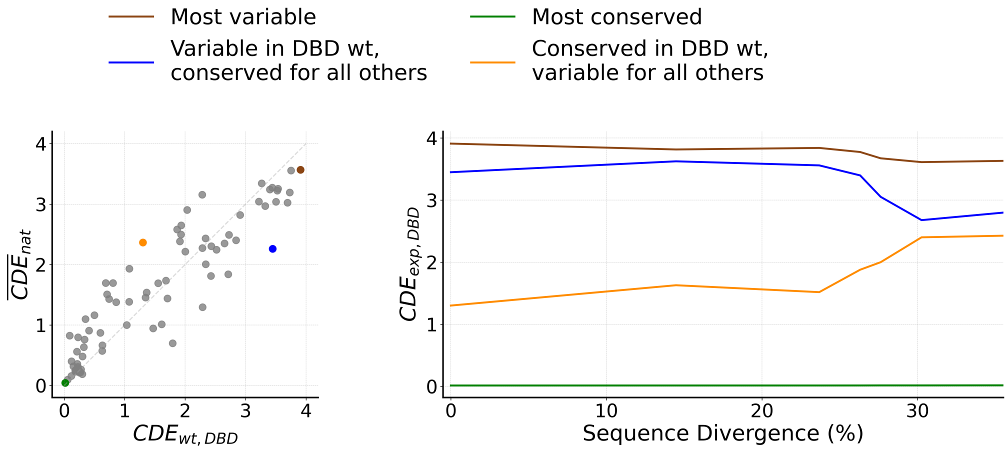

Fig. S7. Evolutionary dynamics of variable, conserved, and epistatic sites for the DBD experimentally studied ancestors. Left: Illustration of four prototypical sites, same figure as on the left of Fig. S4C, repeated here for convenience. Right: Instead of the in silico Monte Carlo chains used on the right of Fig. S4C, we repeat the same calculation for the experimentally investigated DBD sequences, along the reconstructed phylogenetic lineage (seven sequences) starting from the ancestral protein and reaching *C. teleta* SR, i.e. the upper purple branch in Ref. [7, Fig.1B]. We plot the CDE of the four sites identified in the left panel, computed from the bmDCA model, as a function of the sequence divergence, starting from the ancestor (0%) to the *C. teleta* SR (34%). The evolution of the CDE is similar to the in silico trajectories of Fig. S4C.

E. Single-mutation effects and codon bias

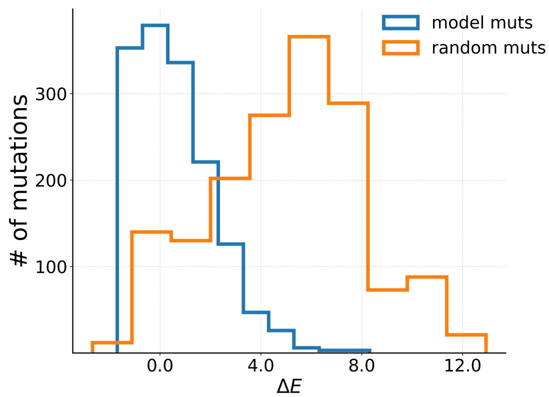

Fig. S8. Neutrality of mutational effects in simulated evolutionary dynamics. Histogram of mutational effects of random point mutations (orange, not taking into account nucleotides) from DBD reconstructed ancestor vs. the histogram of in-silico observed mutations taking place over a sample of short trajectories around the same wt (blue, using the sampling procedure based on nucleotides). Higher ΔE indicates more deleterious mutations and vice-versa.

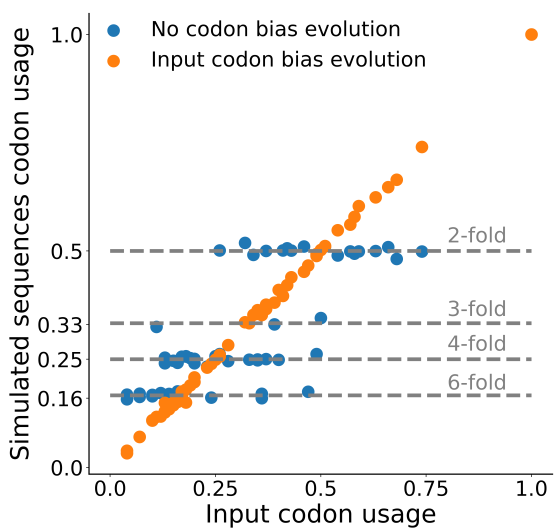

Fig. S9. Effect of the codon bias in sampling algorithm. Scatter plot of E.coli codon usage vs. codon usage of in-silico sequences simulated with the E.coli codon bias (orange) or without codon bias (blue). All simulations were done for the DBD family. As we expect, sequences simulated with E.coli codon bias correctly reproduce E.coli codon usage. In contrast, if no codon bias is introduced, the codon usage resembles the n-fold degeneracies of genetic code: each codon $\mathbf{n}$ has a usage which is $1/[N(A(\mathbf{n}))]$ where $N(A(\mathbf{n}))$ indicates how many codons code the same amino acid of codon $\mathbf{n}$.

F. Role of insertions and deletions

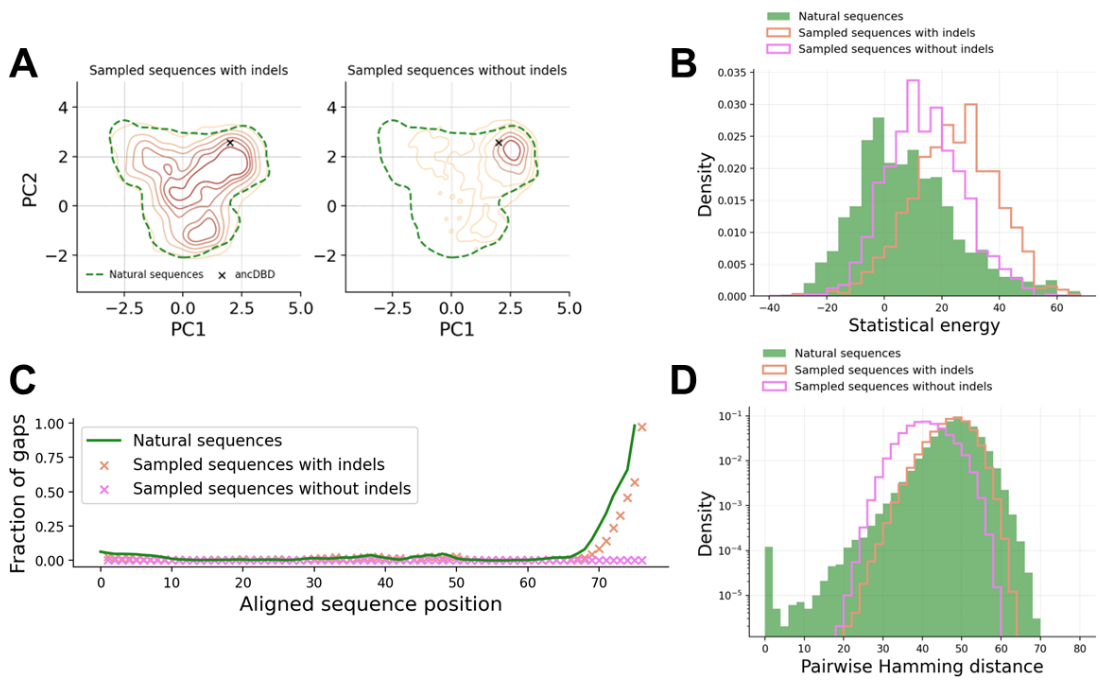

Fig. S10. Necessity of indels to retain generative properties of the sampling algorithm. (A) Left: Projection over the two principal components (PCA) of natural sequences and sequences sampled with the standard algorithm (i.e. with indels). Right: Projection over the two principal components (PCA) of natural sequences and sequences sampled with a modified algorithm that excludes indels. In black the reconstructed ancestral sequence in which simulation has started. (B) Histogram of statistical energies for natural sequences (green), sequences sampled with indels (orange) and without indels (pink). (C) Fraction of gaps at every sequence position for natural sequences (green), sequences sampled with indels (orange) and without indels (pink). (D) Histogram of pairwise hamming distance among natural sequences (green), sequences sampled with indels (orange) and without indels (pink). All simulations were performed for the DBD family.

3. CONTINGENCY AND ENTRENCHMENT: A COMPARISON WITH EXPERIMENTAL DATA

To further validate our evolutionary model, we test our results for contingency and entrenchment against experimental data. The authors of Ref. [7] reconstructed a phylogenetic lineage of a protein belonging to the DNA Binding Domain (DBD) family, from an ancient ancestor down to a present-day sequence, with a sequence divergence of about 35% between the ancestor and the most recent sequence. Then, they experimentally measured the effect of all mutations, i.e. a DMS, around several protein sequences along the lineage, with variable sequence divergence. In line with expectations derived from our model, contingency and entrenchment were experimentally detected in those data. We can then quantitatively compare these experimental results with our in silico results.

First of all, we note that the DCA mutational score is derived from naturally occurring proteins, and it is therefore a generic proxy for the general fitness of a protein along multiple conditions and phenotypes that are relevant to natural selection. Hence, to compare with a specific experiment, we need to map the experimental fitness score ΔF considered in that experiment to our scores ΔE. To do so, following previous work [8, 9], we consider the ensemble of single mutational effects ΔF measured in experiments (i.e., we collect all DMSs from Ref. [7]) and for each mutation (in its context) we compute the corresponding ΔE from the model. We then independently sort both lists of ΔE and ΔF, and plot the sorted lists against each other in Fig. S11C (considering that high fitness corresponds to low statistical energy). In this way, we obtain a monotonous function that estimates the non-linear mapping $\Delta F = \phi(\Delta E)$ and converts energy variations into fitness variations for this particular experiment.

Given $\phi(\Delta E)$, we repeat the analysis of contingency and entrenchment discussed in Fig. S3 as applied to the DBD family, converting the ΔE into ΔF using the non-linear mapping. Also, to allow for a direct comparison with experiments, we plot simulation results as a function of sequence divergence between the sequences at times t_{MUT} and $t + t_{\mathrm{MUT}}$. The in silico results are reported in Fig. S11A for contingency and in Fig. S11B for entrenchment (grey lines and shades). Note that these results are obtained on typical evolutionary trajectories generated by the model, that are uncorrelated from the specific lineage studied experimentally in [7].

Next, we superimpose our in silico results with the experimental measurements of $\Delta\Delta F$ obtained by comparing the effect of the same mutation, taken from any pair of DMS around two distinct wildtypes along the phylogenetic lineage in Ref. [7], as a function of the divergence between these wildtypes. The red points and standard deviations in Figs. S11A-B show, with exception of few outliers, a good quantitative agreement with our in silico results. To complete the picture, we also compute the ΔE from the model around the same wild types used in the experiment, convert them into ΔF using the non-linear mapping ϕ, and perform the same analysis (blue points), obtaining consistent results. We thus see that our evolutionary model captures the trend in experimental data, thus confirming its usefulness as a tool to recapitulate and expand experimental evolution results.

For the sake of completeness, we also compare the single $\Delta\Delta F$ of mutations for experimental and in silico fitness, irrespectively of the distance between the two divergent reconstructed homologs (Fig. S12). We see that our model does not quantitatively predict a $\Delta\Delta F$ given the two sequences and the mutation taking place among them. There are two main motivations for this lack of predictive power. Firstly, as we look more carefully at the scale of $\Delta\Delta F$ (Fig. S12) as compared to that of ΔF (Fig. S11C) we see that contingency and entrenchment are very small effects (in a context of much higher experimental variability) that can be appreciated only at a sequence divergence of $30 - 40\%$ and are consequently pretty hard to model. Secondly, our statistical model is by construction reflecting the global landscape of highly divergent ($70 - 80\%$ of Hamming distance) natural sequences belonging to the same family, hence it cannot have the appropriate resolution for measuring a fine scale change in ΔF over time caused by a single mutation.

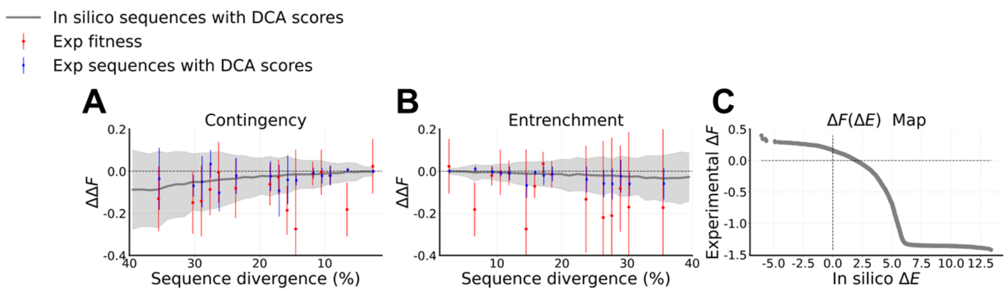

Fig. S11. Comparison between in silico and experimental results for the DBD family. (A) Contingency: amino acids appearing for the first time over an evolutionary trajectory at step t_{MUT} are introduced back in all sequences at $t < t_{\mathrm{MUT}}$ and the difference of mutational effect at t and t_{MUT} is displayed as a function of sequence divergence between the two reference sequences. (B) Entrenchment: amino acids appearing for the last time over an evolutionary trajectory at step t_{MUT} are introduced in all sequences at $t > t_{\mathrm{MUT}}$ and the difference of mutational effect at t and t_{MUT} is displayed as a function of sequence divergence between the two reference sequences. Results in (A) and (B) compare in silico sequences along an ensemble of trajectories generated by our evolutionary model (grey), experimental sequences with a score based on the experimental fitness (red) and experimental sequences with DCA scores based on the in silico model energy (blue). Faded area and error bars are the standard deviation of all mutations happening at that sequence divergence. (C) Visualization of the mapping $\Delta F = \phi(\Delta E)$ that is used to convert DCA energies into experimental fitness, constructed as described in the text.

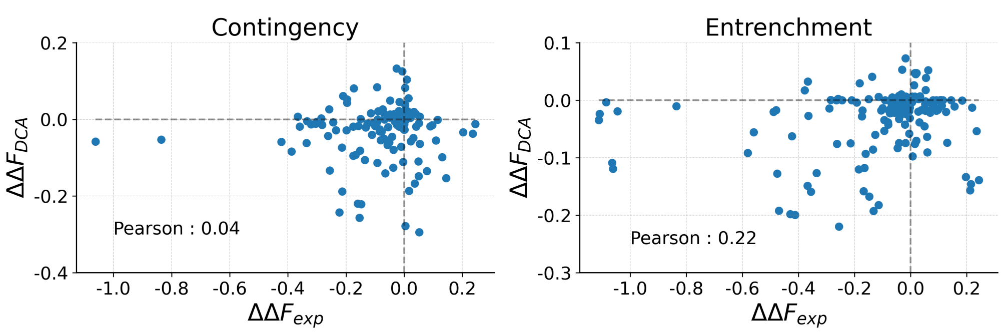

Fig. S12. Comparison between in silico and experimental change in mutational effects over time for the DBD family. We take all the single $\Delta\Delta F_{exp}$ irrespectively of the Hamming distance between the experimental sequences and we compare them with the same changes of mutational effects $\Delta\Delta F_{DCA}$ predicted by the model, showing the Pearson correlation coefficient. We do the analysis separately for changes in mutational effects linked to contingency and entrenchment. The $\Delta\Delta F_{DCA}$ is obtained by the mapping $\Delta F = \phi(\Delta E)$ that is used to convert DCA energies into experimental fitnesses, constructed as described in the text.

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